

Solved selected problems of Classical Electrodynamics - Hans Ohanian

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Chapter 7 - Vector Calculus in Spacetime

Exercises

Solution. Exercise 1.

We see that

$$\begin{aligned}
 a^\alpha{}_\mu \eta_{\alpha\beta} a^\beta{}_\nu &= \begin{pmatrix} \gamma & -V\gamma & 0 & 0 \\ -V\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \gamma & -V\gamma & 0 & 0 \\ -V\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} \gamma & V\gamma & 0 & 0 \\ -V\gamma & -\gamma & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \gamma & -V\gamma & 0 & 0 \\ -V\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} \gamma^2 - V^2\gamma^2 & -V\gamma^2 + V\gamma^2 & 0 & 0 \\ -V\gamma^2 + V\gamma^2 & V^2\gamma^2 - \gamma^2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \\
 &= \begin{pmatrix} \gamma^2(1 - V^2) & 0 & 0 & 0 \\ 0 & \gamma^2(V^2 - 1) & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \\
 &= \eta_{\mu\nu}
 \end{aligned}$$

Where we used that $1 - V^2 = 1/\gamma^2$.

□

Solution. Exercise 2.

Equation (1.38) states that

$$a^{km}a^{kl} = \delta^{ml}$$

Let us define $a^\mu{}_\nu$ as

$$a^\mu{}_\nu = \begin{pmatrix} 1 & 0 \\ 0 & a^k{}_l \end{pmatrix}$$

Where $a^k{}_l$ is a spatial rotation matrix.

Now, using $a^\alpha{}_\mu \eta_{\alpha\beta} a^\beta{}_\nu = \eta_{\mu\nu}$ when $\mu = m$ and $\nu = l$ i.e. we take the spatial indices of the matrix $a^\mu{}_\nu$ we see that

$$\begin{aligned} a^\alpha{}_m \eta_{\alpha\beta} a^\beta{}_l &= a^0{}_m \eta_{00} a^0{}_l + a^1{}_m \eta_{11} a^1{}_l + a^2{}_m \eta_{22} a^2{}_l + a^3{}_m \eta_{33} a^3{}_l \\ &= a^1{}_m \eta_{11} a^1{}_l + a^2{}_m \eta_{22} a^2{}_l + a^3{}_m \eta_{33} a^3{}_l \\ &= a^k{}_m \eta_{kj} a^j{}_l \end{aligned}$$

Where we used that $a^0{}_m$ is 0 when m is a spatial coordinate. Also, we changed α to k and β to j so symbolize that they can only take the spatial coordinate indices.

Therefore, since $\eta_{kj} = -\delta_{kj}$ and $\delta_{kj} a^j{}_l = a^k{}_l$ we have that

$$\begin{aligned} a^k{}_m \eta_{kj} a^j{}_l &= \eta_{ml} \\ -a^k{}_m \delta_{kj} a^j{}_l &= -\delta_{ml} \\ a^k{}_m a^k{}_l &= \delta_{ml} \end{aligned}$$

□

Solution. Exercise 3.

We note that

$$\eta^{\alpha\mu}\eta_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \delta^\alpha{}_\nu$$

Then from equation (13) we know that

$$x_\mu = \eta_{\mu\nu}x^\nu$$

So multiplying both sides by $\eta^{\alpha\mu}$ we get that

$$\eta^{\alpha\mu}x_\mu = \eta^{\alpha\mu}\eta_{\mu\nu}x^\nu = \delta^\alpha{}_\nu x^\nu = x^\alpha$$

If we rename the free index α to ν we get the equation (15)

$$x^\nu = \eta^{\nu\mu}x_\mu$$

Therefore, equation (15) agrees with equation (13). □

Solution. Exercise 4.

Let

$$a^\nu{}_\alpha = \begin{pmatrix} \gamma & -V\gamma & 0 & 0 \\ -V\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Then

$$\begin{aligned} a_\mu{}^\beta &= \eta_{\mu\nu} a^\nu{}_\alpha \eta^{\alpha\beta} \\ &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \gamma & -V\gamma & 0 & 0 \\ -V\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \\ &= \begin{pmatrix} \gamma & -V\gamma & 0 & 0 \\ V\gamma & -\gamma & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \\ &= \begin{pmatrix} \gamma & V\gamma & 0 & 0 \\ V\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{aligned}$$

□

Solution. Exercise 5.

Let us compute $a^\alpha{}_\nu a_\alpha{}^\mu$ as follows

$$\begin{aligned}
a^\alpha{}_\nu a_\alpha{}^\mu &= \begin{pmatrix} \gamma & -V\gamma & 0 & 0 \\ -V\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma & V\gamma & 0 & 0 \\ V\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} \gamma^2 - V^2\gamma^2 & V\gamma^2 - V\gamma^2 & 0 & 0 \\ -V\gamma^2 + V\gamma^2 & -V^2\gamma^2 + \gamma^2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} \gamma^2(1 - V^2) & 0 & 0 & 0 \\ 0 & \gamma^2(1 - V^2) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}
\end{aligned}$$

Also, if we compute $a_\alpha{}^\mu a^\alpha{}_\nu$ we get that

$$\begin{aligned}
a_\alpha{}^\mu a^\alpha{}_\nu &= \begin{pmatrix} \gamma & V\gamma & 0 & 0 \\ V\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma & -V\gamma & 0 & 0 \\ -V\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} \gamma^2 - V^2\gamma^2 & -V\gamma^2 + V\gamma^2 & 0 & 0 \\ V\gamma^2 - V\gamma^2 & -V^2\gamma^2 + \gamma^2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}
\end{aligned}$$

So, in both cases we get the identity matrix, this implies that $a^\alpha{}_\nu$ is the inverse of $a_\alpha{}^\mu$ and viceversa. $\square$

Solution. Exercise 6.

Equation (9) states that

$$a^\alpha{}_\mu \eta_{\alpha\beta} a^\beta{}_\nu = \eta_{\nu\mu}$$

Then multiplying the equation by $\eta^{\nu\sigma}$ we get that

$$a^\alpha{}_\mu \eta_{\alpha\beta} a^\beta{}_\nu \eta^{\nu\sigma} = \eta_{\nu\mu} \eta^{\nu\sigma}$$

We proved in Exercise 3 that $\eta_{\nu\mu} \eta^{\nu\sigma} = \delta^\sigma{}_\mu$ and we know because of equation (20) that

$$\eta_{\alpha\beta} a^\beta{}_\nu \eta^{\nu\sigma} = a_\alpha{}^\sigma$$

So we get that

$$a^\alpha{}_\mu a_\alpha{}^\sigma = \delta^\sigma{}_\mu$$

Or

$$a_\alpha{}^\mu a^\alpha{}_\nu = \delta^\mu{}_\nu$$

Where we used that $a^\alpha{}_\mu$ and $a_\alpha{}^\sigma$ are inverses of each other so they commute as we proved in Exercise 5 and we renamed the indexes $\sigma \rightarrow \mu$ and $\mu \rightarrow \nu$. $\square$

Solution. Exercise 7.

Let us consider the product $x^\mu x^\nu$. Since the position vectors transform as

$$x'^\mu = a^\mu_\alpha x^\alpha$$

Then we can write that

$$x'^\mu x'^\nu = a^\mu_\alpha x^\alpha a^\nu_\beta x^\beta = a^\mu_\alpha a^\nu_\beta x^\alpha x^\beta$$

So the product $x^\mu x^\nu$ transforms as a tensor, and hence it's a tensor. $\square$

Solution. Exercise 8.

We note that

$$a^\beta_\nu = a^\alpha_\nu \delta_\alpha^\beta$$

From equation (23) we have that

$$a_\beta^\mu a^\beta_\nu = \delta^\mu_\nu$$

So we can write that

$$a_\beta^\mu a^\alpha_\nu \delta_\alpha^\beta = \delta^\mu_\nu$$

Therefore δ^μ_ν transforms as a tensor and hence it's a tensor. $\square$

Solution. Exercise 9.

From equation (9) we know that

$$\eta_{\alpha\beta} = a^\mu_\alpha \eta_{\mu\nu} a^\nu_\beta$$

So multiplying the equation by $a_\rho^\alpha a_\sigma^\beta$ we get that

$$a_\rho^\alpha a_\sigma^\beta \eta_{\alpha\beta} = a_\rho^\alpha a^\mu_\alpha \eta_{\mu\nu} a^\nu_\beta a_\sigma^\beta = \delta_\rho^\mu \eta_{\mu\nu} \delta_\sigma^\nu$$

Therefore

$$\eta_{\rho\sigma} = a_\rho^\alpha a_\sigma^\beta \eta_{\alpha\beta}$$

So $\eta_{\mu\nu}$ transforms as a tensor. $\square$

Solution. Exercise 10.

We know that

$$A'^{\mu} = a^{\mu}_{\alpha} A^{\alpha} \quad \text{and that} \quad B'_{\mu} = a_{\mu}^{\beta} B_{\beta}$$

Then

$$A'^{\mu} B'_{\mu} = a^{\mu}_{\alpha} A^{\alpha} a_{\mu}^{\beta} B_{\beta} = a^{\mu}_{\alpha} a_{\mu}^{\beta} A^{\alpha} B_{\beta} = \delta_{\alpha}^{\beta} A^{\alpha} B_{\beta} = A^{\beta} B_{\beta}$$

□

Solution. Exercise 11.

Let B^{μ} be a vector and $A_{\alpha\mu}$ be a tensor then

$$\begin{aligned} A'_{\alpha\mu} B'^{\mu} &= a_{\alpha}^{\sigma} a_{\mu}^{\beta} A_{\sigma\beta} a^{\mu}_{\rho} B^{\rho} \\ &= a_{\alpha}^{\sigma} a_{\mu}^{\beta} a^{\mu}_{\rho} A_{\sigma\beta} B^{\rho} \\ &= a_{\alpha}^{\sigma} \delta_{\rho}^{\beta} A_{\sigma\beta} B^{\rho} \\ &= a_{\alpha}^{\sigma} A_{\sigma\beta} B^{\beta} \end{aligned}$$

Where we used the laws of transformation of B^{μ} and $A_{\alpha\mu}$, and we see that $A_{\alpha\mu} B^{\mu}$ transforms as a vector. □

Solution. Exercise 12.

Let $T^{\mu\nu}$ be a tensor, we see that

$$T'^{\mu}_{\nu} = a^{\mu}_{\alpha} a_{\nu}^{\beta} T^{\alpha}_{\beta}$$

If we let $\nu = \mu$ then we have that

$$T'^{\mu}_{\mu} = a^{\mu}_{\alpha} a_{\mu}^{\beta} T^{\alpha}_{\beta} = \delta_{\alpha}^{\beta} T^{\alpha}_{\beta} = T^{\alpha}_{\alpha}$$

We see that T^{μ}_{μ} remains unchanged under Lorentz transformations so it is a scalar. □

Solution. Exercise 13.

Let us compute $u_\mu u^\mu$ as follows

$$\begin{aligned}
u_\mu u^\mu &= \eta_{\mu\nu} u^\nu u^\mu \\
&= \frac{c^2}{1 - v^2/c^2} - \frac{(dx/dt)^2}{1 - v^2/c^2} - \frac{(dy/dt)^2}{1 - v^2/c^2} - \frac{(dz/dt)^2}{1 - v^2/c^2} \\
&= \frac{c^2}{1 - v^2/c^2} - \frac{v^2}{1 - v^2/c^2} \\
&= \frac{c^2 - v^2}{1 - v^2/c^2} \\
&= \frac{c^2(1 - v^2/c^2)}{1 - v^2/c^2} \\
&= c^2
\end{aligned}$$

Where we used that $v^2 = (dx/dt)^2 + (dy/dt)^2 + (dz/dt)^2$. □

Solution. Exercise 14.

We know that $u_\mu u^\mu = c^2$ and that $p^\mu = m u^\mu$ so

$$p_\mu p^\mu = m^2 u_\mu u^\mu = m^2 c^2$$

Therefore

$$p_\mu p^\mu = \frac{E^2}{c^2} - p_x^2 - p_y^2 - p_z^2 = m^2 c^2$$

□

Solution. Exercise 15.

We know from the Lorentz transformations that

$$\begin{aligned} E' &= \frac{E - VP_x}{\sqrt{1 - V^2/c^2}} & P'_x &= \frac{P_x - VE/c^2}{\sqrt{1 - V^2/c^2}} \\ P'_y &= P_y & P'_z &= P_z \end{aligned}$$

Where the primed quantities are the energy and momentum measured in a frame S' which is moving with velocity V with respect to an inertial frame S . In our case, we get that $E_0 = E'$ so

$$E - VP_x = E_0 \sqrt{1 - V^2/c^2}$$

Also, in our case, in S' we have that $P'_x = P'_y = P'_z = 0$ so we get that

$$P_x = VE/c^2 \quad P_y = P_z = 0$$

Therefore

$$\begin{aligned} E(1 - V^2/c^2) &= E_0 \sqrt{1 - V^2/c^2} \\ E &= E_0 \frac{\sqrt{1 - V^2/c^2}}{1 - V^2/c^2} \\ E &= \frac{E_0}{\sqrt{1 - V^2/c^2}} \end{aligned}$$

Replacing E we get that

$$P_x = \frac{E_0 V/c^2}{\sqrt{1 - V^2/c^2}}$$

Hence

$$\mathbf{P} = \frac{E_0 \mathbf{V}/c^2}{\sqrt{1 - V^2/c^2}}$$

□

Solution. Exercise 16.

Let $\Phi(t, x) = f(ct + x)$ and let us call $v = ct + x$ then, we get that

$$\begin{aligned}\frac{\partial \Phi}{\partial t} &= \frac{\partial f(v)}{\partial t} = \frac{\partial f(v)}{\partial v} \frac{\partial v}{\partial t} = c \frac{\partial f(v)}{\partial v} \\ \frac{\partial \Phi}{\partial x} &= \frac{\partial f(v)}{\partial x} = \frac{\partial f(v)}{\partial v} \frac{\partial v}{\partial x} = \frac{\partial f(v)}{\partial v}\end{aligned}$$

Differentiating again we have that

$$\begin{aligned}\frac{\partial^2 \Phi}{\partial t^2} &= \frac{\partial}{\partial t} \left(c \frac{\partial f(v)}{\partial v} \right) = c \frac{\partial}{\partial v} \left(\frac{\partial f(v)}{\partial t} \right) = c^2 \frac{\partial^2 f(v)}{\partial v^2} \\ \frac{\partial^2 \Phi}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial f(v)}{\partial v} \right) = \frac{\partial}{\partial v} \left(\frac{\partial f(v)}{\partial x} \right) = \frac{\partial^2 f(v)}{\partial v^2}\end{aligned}$$

Then

$$\frac{1}{c^2} \frac{\partial^2 \Phi}{\partial t^2} - \frac{\partial^2 \Phi}{\partial x^2} = \frac{1}{c^2} \left(c^2 \frac{\partial^2 f(v)}{\partial v^2} \right) - \frac{\partial^2 f(v)}{\partial v^2} = 0$$

Therefore $f(ct + x)$ is a solution to equation (96). If we try $f(ct - x)$ with $v = ct - x$ we see that

$$\frac{\partial \Phi}{\partial x} = \frac{\partial f(v)}{\partial x} = \frac{\partial f(v)}{\partial v} \frac{\partial v}{\partial x} = - \frac{\partial f(v)}{\partial v}$$

and

$$\frac{\partial^2 \Phi}{\partial x^2} = \frac{\partial}{\partial x} \left(- \frac{\partial f(v)}{\partial v} \right) = \frac{\partial}{\partial v} \left(- \frac{\partial f(v)}{\partial x} \right) = \frac{\partial}{\partial v} \left(- \left(- \frac{\partial f(v)}{\partial v} \right) \right) = \frac{\partial^2 f(v)}{\partial v^2}$$

This is the same result we get for $f(ct + x)$, therefore equation (96) is also satisfied by $f(ct - x)$. □

Solution. Exercise 17.

From the Lorentz-transformation equations for the current four-vector, we know that

$$\rho' = \frac{\rho - Vj_x/c^2}{\sqrt{1 - V^2/c^2}} \quad j'_x = \frac{j_x - V\rho}{\sqrt{1 - V^2/c^2}}$$

If a wire along the x axis is electrically neutral but carries a current in the xyz frame then $\rho = 0$ and $j_x \neq 0$ then

$$\rho' = -\frac{Vj_x/c^2}{\sqrt{1 - V^2/c^2}} \quad j'_x = \frac{j_x}{\sqrt{1 - V^2/c^2}}$$

Since $\rho' \neq 0$ this implies that the wire will not be neutral in the $x'y'z'$ frame. $\square$